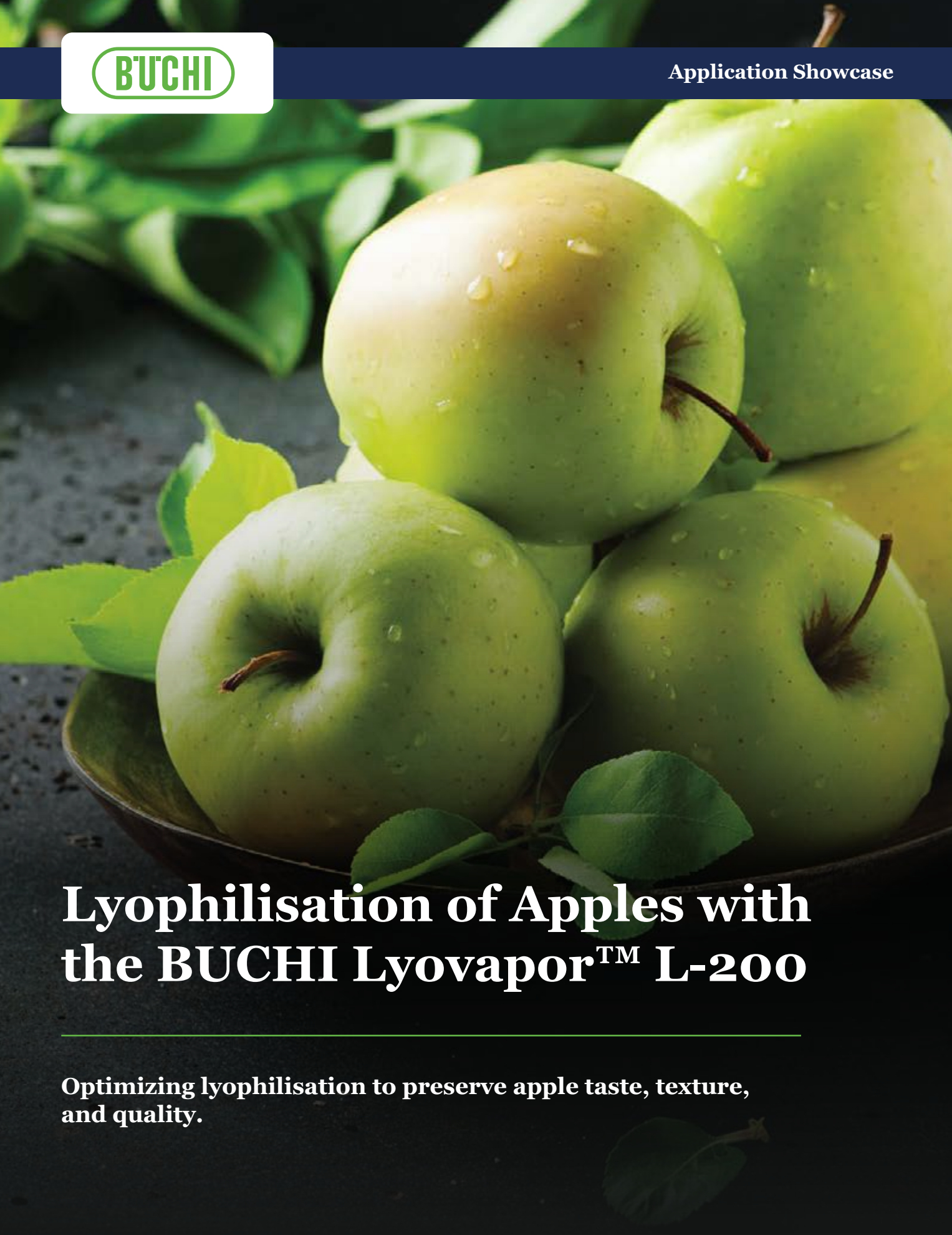


The background of the entire page is a high-quality photograph of several bright green apples. The apples are covered in fine water droplets, giving them a fresh and crisp appearance. They are arranged in a dark, rustic-style bowl, with some green leaves scattered around them. The lighting is soft, highlighting the texture of the apple skin and the glistening water droplets.

Lyophilisation of Apples with the BUCHI Lyovapor™ L-200

Optimizing lyophilisation to preserve apple taste, texture,
and quality.

Preservation of seasonal fruits for long-term consumption, storage by freeze drying is a common approach aiming on maintaining the original taste, appearance, texture. Here, we developed a freeze-drying method with optimized process time and residual moisture content for apples.

Keywords or phrases:

Freeze drying, Lyophilisation, Moisture analysis, Aroma retention, Apple preservation, Sublimation process

INTRODUCTION

The taste of apples is partly determined by the biogenesis of numerous volatile¹ compounds. More than 300 compounds from various crops of apples have been identified as aromatics. However, only a few compounds emitted from apples have been determined to have a decisive impact on sensory quality, so they have been designated as impact compounds in apples.

A considerable portion of these compounds can evaporate during the drying process, resulting in bad taste, which varies between different drying methods and conditions. Quantitative knowledge of the losses of aromatic compounds during drying fruits is also essential to improve the recovery of substances in various commercial activities. Several factors influence the retention of aroma, including the vapour pressure of each compound, its relative volatility, the drying temperature, and the composition of the food product.

Conventional drying methods such as hot air drying, vacuum oven drying, and sun drying, have the disadvantages of reducing the quality and taste of the original food due to loss

of heat-sensitive substances, long drying time depending on weather conditions, and sterility issues². On the other hand, freeze drying is one of the most sophisticated dehydration methods. Freeze drying is a gentle way to remove moisture without disturbing the internal structure of the sample.

Freeze drying is a sublimation process where solid particle is directly converted into vapour form without coming into liquid form. Another important advantage is that freeze drying is accomplished at relatively low temperatures at which fragile compounds are preserved.

Freeze-dried foods are mainly considered of higher quality than other dehydrated products because they can be reconstituted rapidly to products that closely resemble the original taste.

It provides dried products of porous structure, no shrinkage, retention of morphology, superior taste retention³, surface area, high yield, long shelf life, high viability, weight reduction, the possibility of sterile handling and better rehydration properties. However, its advantages are directly weighed against its associated high treatment cost.

EQUIPMENT

- ▶ BUCHI Lyovapor™ L-200 Pro.
- ▶ BUCHI Lyovapor™ Software.
- ▶ Vacuum pump Pfeiffer Duo 6".
- ▶ Heated Shelves.
- ▶ Stainless Steel trays.
- ▶ Freezer -40 °C – Haier.
- ▶ Balance – Mettler Toledo.
- ▶ Karl Fischer titrator moisture analyzer – Mettler Toledo.

MATERIALS

- ▶ Fresh red apples, purchased from a local supermarket.
- ▶ Knife.
- ▶ Lime juice with water.

EXPERIMENTAL

Sample preparation

Apples contain high levels of an enzyme called polyphenol oxidase. When an apple is sliced open, the cells are ruptured. With the help of oxygen in the air around damaged cells, polyphenol oxidase initiates a series of chemical reactions, transforming polyphenols and eventually forming melanin-brown pigments⁵. Lemon contains ascorbic acid, which not only reduces pH (as does the citric acid also found in lemons) but also the oxidative reaction can be reversed by reduction. The brown colour can't be avoided completely, but it had a better appearance than un used one.

Apples have been cut into pieces approximately 3 mm in width and soaked in lemon juice water for 10 minutes and placed into a tray.

The sample temperature sensor is placed in apple slices to monitor the sample temperature during the freeze drying process. The product trays are kept in deep freezer for overnight freezing at a temperature of -30°C .

Lyovapor™ L-200 settings

After 12 hours of deep freezing, the slices were completely frozen. The frozen trays and frozen shelf have been transferred into the Lyovapor L-200 instrument with a PMMA drying chamber at ambient pressure.

The set method was designed considering the critical parameters such as triple temperature at -5°C , triple point pressure at 0.5 mbar and critical temperature at 55°C ⁴.

The freeze drying process was initiated with the loading step at which shelf temperature needs to be set, e.g. -25°C due to the freezing in the freezer. By keeping the shelf at the same temperature as samples, sample melting due to heat transfer will be minimized and even drying can be ensured. In the drying process, the shelf temperature was gradually increased from -10°C to 40°C to facilitate the sublimation of water from the sample. To keep with the optimal sublimation

and prevent the collapse of the sample, the pressure was maintained at a constant level of 0.300 mbar throughout the primary drying process.

In the secondary drying phase, the pressure was minimized to achieve a low moisture content in the final product. This step is critical to achieve the stability and product shelf life by removing any residual moisture by desorption that could cause degradation or spoilage over time. Details can be found in Figure 2.

During this run, the Lyovapor monitoring app is used as an assisting tool to monitor the overnight lyophilization process. The app helps to keep a close check on the progress of the process and promptly alerts the user in case of any discrepancies in set parameters.

Step	1	2	3	4	5	6
Phase	Loading	Primary drying	Primary drying	Primary drying	Secondary drying	Secondary drying
Duration	0:00	0:00	0:00	0:00	0:00	0:00
Shelf temperature	-25	-25	-25	-25	-25	-25
Shelf temperature gradient	0.0	0.0	0.0	0.0	0.0	0.0
Pressure zone	Backfilled	Backfilled	Backfilled	Backfilled	Minimum	Minimum
Pressure	0.400	0.400	0.400	0.400		
Safety pressure limit	1.000	1.000	1.000	1.000		
Safety pressure duration	10	10	10	10		
Procession Status						
Primary drying phase time	17:00					
Secondary drying phase time	00:00					
Total time	24:00					

Figure 1: Parameters of the primary and secondary drying phase set on the Lyovapor™ Software.

RESULT AND DISCUSSION

Run Data Plot

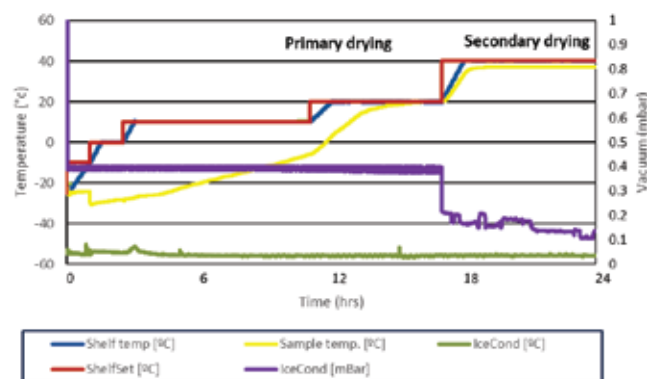


Figure 2: Run data of the primary and secondary drying steps plotting from the Lyovapor™ software.

The figure 1 depicts the process of a freezing drying cycle. After the completion of 17 hours, the sample temperature reaches the shelf temperature of 20 °C. This convergence is a significant indication that the primary drying phase has reached the end. During the primary drying phase, the bulk water is removed from the sample by sublimation which takes lots of energy. The transition into the secondary drying phase begins at 18 hours. In the secondary drying phase, the sample is dried by removing bounded moisture. The amount of this water is rather low and therefore the sample follows the set shelf temperature quickly.

Since freeze-dried products exhibit highly hygroscopic nature in general, the dry apple slices after the freeze drying process are immediately transferred to a sealed or closed container to avoid moisture absorption from the surrounding atmosphere.

PRODUCT APPEARANCE



Figure 3: Apple slices before (left) and after (right) freeze drying.

The appearance did change slightly before and after the freeze drying. But the consistency can be classified as dry matter. The initial total weight of the sample was 1982 grams. The total weight is drastically reduced to 518 grams after drying. This result goes along with the proposed water content of over 80 % in fruit in the introduction part.

RESIDUAL MOISTURE ANALYSIS

The freeze-dried apples were analyzed for moisture content by Karl Fischer Titration.

Description	Result
Actual apple weight (without moisture)	502 gr
Moisture content in dry sample	2.64%
Total removed moisture	97%

Table 1: Residual Moisture results of freeze drying apple slices.

CONCLUSION

With the Lyovapor™ L-200, drying has been completed with 97 % of the moisture removed in 24 h which shows a high drying efficiency. The final dry fruit are dried without changing their appearance, texture and colour with a moisture content of < 2.7 %.

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